UNIVERSITÀ DI PISA

Department of Computer Science

Master's in Data Science and Business Informatics

Distributed Data Analysis Mining Project Report - Credit Risk Analysis and Loan Default Prediction Using LendingClub Loan Data

LendingClub

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Table of Contents

1 Introduction

Peer-to-peer lending platforms have transformed the way individuals access credit by directly connecting borrowers and investors through online marketplaces. One of the largest platforms in this domain is **LendingClub**, which provides personal loans based on borrowers' financial and credit profiles. While these platforms increase accessibility to credit, they also introduce financial risk because some borrowers may fail to repay their loans. For lenders and investors, accurately assessing borrower risk is therefore essential in order to reduce potential financial losses and support better lending decisions.

With the growing availability of large financial datasets, data mining and machine learning techniques can be applied to analyze borrower behavior and identify patterns related to credit risk. Lending platforms evaluate loan applications using various attributes such as credit score, income, employment history and existing financial obligations. By analyzing these variables, it is possible to gain insights into the factors that influence loan performance and borrower repayment behavior.

This project applies data analysis and machine learning techniques to the LendingClub loan dataset in order to analyze borrower characteristics and explore patterns in lending data.

1.1 Research Questions and Objectives

The main objective of this project is to analyze the LendingClub loan dataset in order to understand borrower characteristics and investigate patterns related to loan repayment behaviour. Data exploration and preprocessing techniques are applied to prepare the dataset for machine learning analysis.
The study focuses on the following research questions:

1. **Can we predict whether a loan will default or not? (classification)**
   This task involves building a classification model that distinguishes between loans that were fully repaid and those that resulted in default.
2. **How does LendingClub determine loan interest rates based on borrower characteristics? (regression)**
   This question explores whether financial indicators such as income, loan amount, interest rate and credit history are associated with the likelihood of loan default.
3. **Can natural borrower risk segments be identified using unsupervised learning? (clustering)**
   Through exploratory data analysis, the project investigates distributions, relationships between variables and potential indicators of credit risk.

By addressing these questions, the project aims to demonstrate how data mining techniques can be used to analyse financial datasets and extract meaningful insights about borrower risk.

2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is conducted to understand the structure and main characteristics of the dataset before applying machine learning techniques. This step examines variable distributions, identifies potential data quality issues and explores relationships between borrower characteristics and loan performance.
The dataset contains loan records issued through the LendingClub platform between **2007 and 2018.** Each observation represents an individual loan and includes borrower financial information, credit history indicators, loan characteristics and repayment outcomes. Through exploratory analysis, we aim to identify patterns related to borrower risk and factors that may influence loan default and interest rates.

2.1 Dataset Structure

The LendingClub dataset contains **2,260,701** loan records described by **151** variables, where each observation represents an individual loan issued through the platform.
The dataset includes both numerical and categorical variables describing borrower characteristics and loan attributes. Numerical variables capture financial indicators such as loan amount, interest rate, income and credit utilization, while categorical variables include borrower profile information such as loan grade, home ownership, loan purpose and verification status.

A summary of the dataset structure is presented in **Table 1**.

Table 1: Dataset Overview

Property	**Value**
Number of observations	2,260,701
Number of variables	151
Numerical variables	112
Categorical variables	39
Observation unit	Individual loan

2.2 Loan Outcomes, Credit Grades and Interest Rates

The LendingClub dataset includes several variables related to loan performance and borrower risk, including the repayment outcome (loan_status), loan interest rate (int_rate) and borrower credit grade (grade). These variables provide insights into how borrower risk characteristics relate to loan outcomes and pricing.

Figure 1 shows the distribution of loan outcomes. The majority of loans fall into the **Fully Paid** or **Current** categories, indicating successful or ongoing repayment, while a smaller portion of loans are **Charged Off**, representing borrower default. This imbalance between default and non-default loans suggests that the classification task may involve class imbalance, where metrics such as **recall and F1-score** are more informative than accuracy. Only **33** observations contain missing values in loan_status, representing a negligible proportion of the dataset and there were also 33 null values in int_rate.

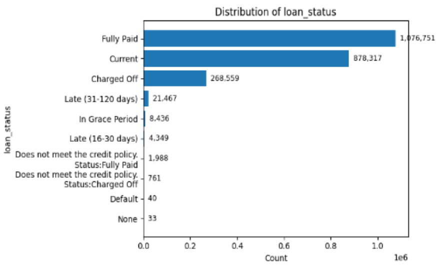

Figure 1: Distribution of loan_status categories.

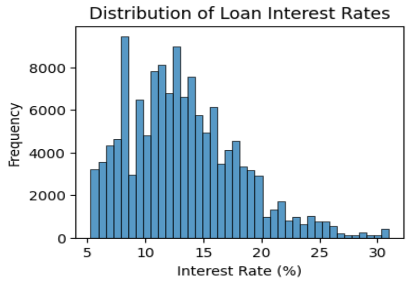

Figure 2: Distribution of loan interest rates.

Figure 2 presents the distribution of loan interest rates. Most loans are issued at moderate interest rates, with fewer loans appearing at the highest rate levels. This variation reflects differences in borrower risk profiles. It also suggests that interest rate is strongly linked to borrower risk and will be an important variable for our regression task.

Borrower risk is further captured by the loan grade, which ranges from A (lowest risk) to G (highest risk). **Figure 3** shows that default probability increases steadily from grade A to grade G, indicating that higher-risk borrower categories experience substantially higher default rates.

Loan grades also strongly influence loan pricing. **Figure 4** shows, interest rates increase systematically across grades, with lower-risk borrowers receiving lower interest rates and higher-risk borrowers facing higher borrowing costs. Although the median interest rate increases across grades, some overlap between adjacent categories suggests that additional borrower characteristics may also influence loan pricing.

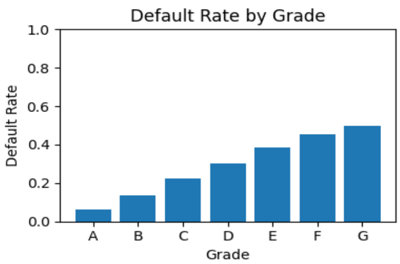

Figure 3: Default rate by loan grade.

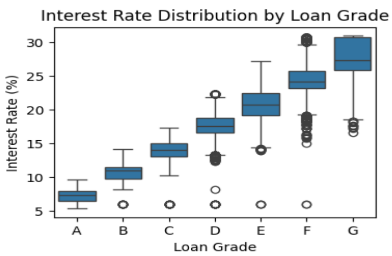

Figure 4: Interest rate distribution across loan grades.

2.3 Missing Value Analysis

Real-world financial datasets often contain missing values because some borrower attributes are recorded only under specific conditions. Therefore, analyzing missing value patterns is an important step before preparing the dataset for modeling. After applying the outcome filtering used for the classification task, the dataset contains **1,348,099** loan records. Examination of missing values shows that several variables contain extremely high proportions of missing entries, with some features exceeding **98-99%** missing values. **Table 2** presents examples of variables with the highest missing percentages.

Table 2: Example variables with highest missing percentages

column	null_count	null_pct
sec_app_mths_since_last_major_derog	1341450	0.9950678696445884
sec_app_revol_util	1329791	0.9864193950147578
revol_bal_joint	1329464	0.9861768312267868
verification_status_joint	1322498	0.9810095549362473
dti_joint	1322296	0.9808597143088156

Most variables with very high missingness are related to secondary applicants or joint loan information, such as sec_app_* features and variables like annual_inc_joint and dti_joint. These fields are recorded only when a loan application includes a co-borrower, which occurs relatively infrequently.

Several delinquency-related variables (e.g., mths_since_last_record) also contain substantial missing values, which often indicate that no delinquency event occurred rather than missing data. Strategies for handling missing values are described later in the Data Preprocessing section.

2.4 Feature Distribution

Given the large number of variables in the dataset, a subset of representative financial features was examined to illustrate the distribution of key borrower and loan characteristics, including loan_amnt, annual_inc, dti and revol_util.

Figure 5 presents the distributions of these variables. Several financial variables exhibit right-skewed distributions, which is common in financial datasets where most observations fall within moderate ranges while a smaller number of observations take substantially larger values. For example, annual_inc shows a pronounced right tail, while loan_amnt is concentrated in lower to mid loan ranges. The distributions of dti and revol_util indicate that most borrowers maintain moderate debt-to-income ratios and credit utilization levels, although some extreme values are present. Such skewness and extreme observations are typical in financial data. In addition to the displayed distributions, numerical variables were visually inspected using distribution plots and boxplots to identify skewness and potential outliers. Categorical feature distributions were also examined, revealing some class imbalance. These patterns were retained because they reflect genuine borrower characteristics rather than data quality issues and were addressed later during preprocessing.

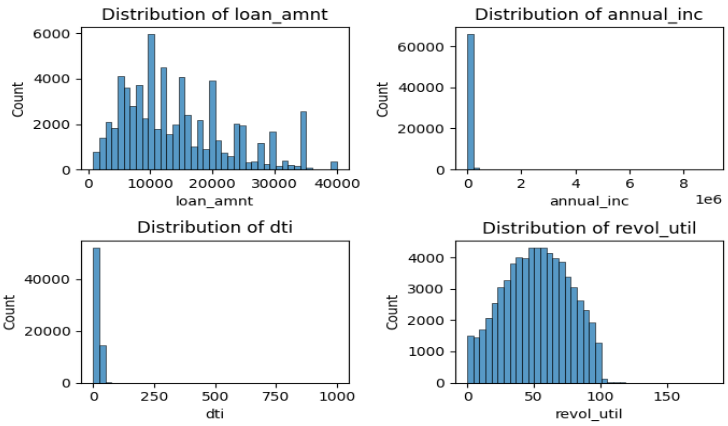

Figure 5: Distribution of a few financial variables.

3 Data Preprocessing

This section describes the steps used to transform the raw LendingClub dataset into a clean and model-ready format. All transformations were applied using PySpark to support scalable processing on the large dataset.

3.1 Outcome Filtering and Binary Target Definition (Classification)

The original dataset contains multiple loan outcome categories in the loan_status variable, including final outcomes (e.g., *Fully Paid*, *Charged Off*) and intermediate states (e.g., *Current*, *Late*, *In Grace Period*). Since the classification task aims to predict whether a loan ultimately defaults, only final loan outcomes were retained, while intermediate states were excluded.

Records with missing values in the required fields (loan_status and int_rate) were first removed. The number of missing observations was negligible (**33** records for each variable, corresponding to **0.0015%** of the dataset).

The remaining loan statuses were mapped into a binary target variable:

- **label = 0 (non-default):** Fully Paid, Does not meet the credit policy - Status: Fully Paid
- **label = 1 (default):** Charged Off, Default, Does not meet the credit policy - Status: Charged Off

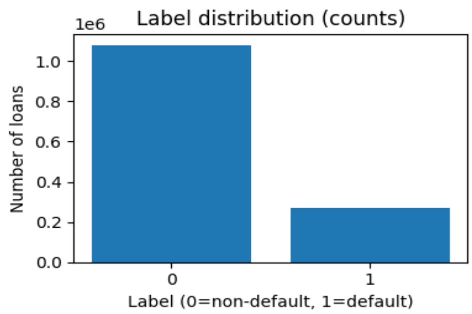

Figure 6: Distribution of binary loan default labels (0 = non-default, 1 = default).

After filtering and target mapping, the dataset was reduced from **2,260,701 to 1,348,099** loan records. As shown in **Figure 6**, the data includes **1,078,739** non-default and **269,360** default loans, corresponding to a default rate of about **20%**, indicating moderate class imbalance.

3.2 Removal of Leakage, Irrelevant Features and Duplicate Check

After defining the classification target, features that could introduce data leakage, as well as irrelevant or unsuitable variables, were removed from the dataset. Data leakage occurs when predictors contain information that would not be available at the time of prediction, leading to overly optimistic model performance.

Several variables in the LendingClub dataset capture events occurring after loan issuance, such as payment history, recoveries and updated account information. Since these variables reflect the repayment process rather than borrower characteristics at loan approval, they were removed prior to modeling. Examples include total_pymnt, total_rec_prncp, recoveries, collection_recovery_fee, last_pymnt_amnt and last_credit_pull_d.

Additional post-issue variables related to borrower hardship or settlement events were also removed. These include columns with prefixes such as hardship_*, settlement_* and debt_settlement_*, which describe repayment difficulties occurring after the loan was issued. This prefix-based filtering removed **19** columns. A few additional post-issue variables not captured by this rule, including deferral_term, payment_plan_start_date and orig_projected_additional_accrued_interest, were also excluded.

Several variables were removed because they contained unstructured text or high-cardinality identifiers that are not suitable for predictive modeling. These include member_id, url, desc, title, emp_title and zip_code.

The variable id was temporarily retained to verify dataset integrity and detect potential duplicate records. A duplicate check confirmed that no duplicated entries were present.

Finally, the variable sub_grade was removed because it acts as a proxy for the broader grade variable and retaining both would introduce redundancy. In total, **44** columns were removed, reducing the feature set from **151** to **107** variables while maintaining **1,348,099** loan observations.

3.3 Type Conversion and Feature Engineering

Several variables in the LendingClub dataset were originally stored as text and required conversion into structured numerical representations suitable for machine learning models. In addition, new features were engineered to better capture relationships between borrower characteristics and loan attributes.

First, several date variables (issue_d, earliest_cr_line and sec_app_earliest_cr_line) were converted into standardized date formats. Using these values, a new feature credit_hist_months was created to represent the length of a borrower's credit history, calculated as the number of months between the loan issue date and the earliest recorded credit line. The parsed issue_date variable was retained temporarily to allow chronological dataset splitting and was removed before final model training.

Several categorical variables were also converted into numerical form. The loan term variable (term) was transformed into term_months and the employment length variable (emp_length) was converted into emp_length_years. In addition, the borrower's credit score was summarized by averaging the lower and upper bounds of the FICO score range (fico_range_low and fico_range_high) to create a single continuous feature fico_avg. To better represent borrower affordability, two ratio-based variables were engineered: loan_income_ratio, representing the loan amount relative to annual income and **installment_income_ratio**, representing the monthly installment relative to borrower income.

The relevance of these variables is reflected in borrower risk patterns. As shown in **Figure 7**, loans with a **60-month term** exhibit substantially higher default rates than loans with **36-month terms**. Similarly, **Figure 8** shows that default probability decreases as the borrower's FICO score increases, highlighting the importance of credit quality in predicting loan performance. After generating these features, several original columns were removed to avoid redundancy, including the original date strings, categorical text variables (term, emp_length) and the FICO score bounds. Core financial variables such as loan_amnt, annual_inc and installment were retained to preserve absolute-scale financial information.

Overall, these transformations produced structured numerical features representing borrower credit history, financial capacity and repayment burden, improving the dataset's suitability for predictive modeling.

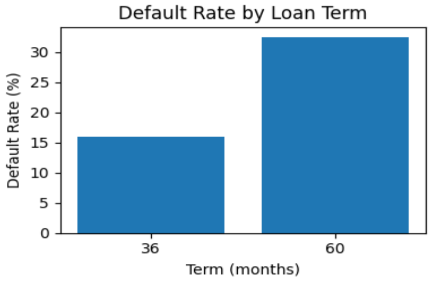

Figure 7: Default rate by loan term.

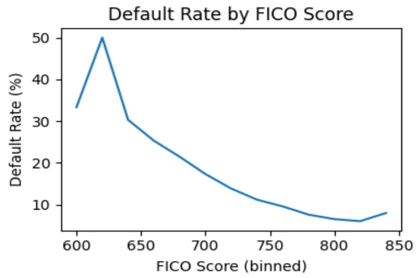

Figure 8: Default rate by borrower FICO score.

3.4 Missingness Filtering, Constant Feature Removal and Validity Constraints

Following the exploratory analysis, additional data quality checks were performed to address sparse features, constant variables and implausible numeric values.

First, columns with extremely high levels of missing values were removed using a threshold of **85%** missingness. Features exceeding this threshold provide limited information for modeling and may introduce unnecessary sparsity. Applying this rule resulted in the removal of **15** variables, many of which were related to secondary applicants or joint loan information (e.g., sec_app_*, annual_inc_joint and dti_joint), which are only recorded when a loan includes a co-borrower.

Next, features with no variation were removed because they do not contribute predictive information. In this dataset, the numeric feature policy_code was identified as constant and was removed. Additionally, categorical variables were examined for extreme class imbalance. Variables where a single category accounted for more than **99.5%** of observations were treated as near-constant and removed. Based on this criterion, the variable pymnt_plan was excluded.

Finally, a set of validity constraints and plausibility bounds was applied to ensure that numerical values remained within realistic ranges. Instead of removing observations, implausible values were corrected or capped. For example, the loan-to-income ratio (loan_income_ratio) was capped at a maximum value of **10**, while the installment-to-income ratio (installment_income_ratio) was capped at **1** to prevent unrealistic ratios caused by extremely small reported incomes. Negative values in variables such as dti and revol_util were corrected to zero and utilization values above **100%** were capped. Credit scores outside the standard FICO range **(300-850)** were treated as missing values and negative credit history lengths were removed.

During exploratory analysis, distribution plots and boxplots revealed the presence of extreme values in several numerical variables. Since financial datasets often contain legitimate high or low values reflecting real borrower behavior, these observations were not removed. Instead, only clearly implausible values were corrected or capped while genuine extreme values were retained. **Figure 9** shows the distribution of the loan-to-income ratio after applying these plausibility constraints. Similarly, categorical variables with imbalanced distributions were preserved because they represent real borrower characteristics rather than data errors.

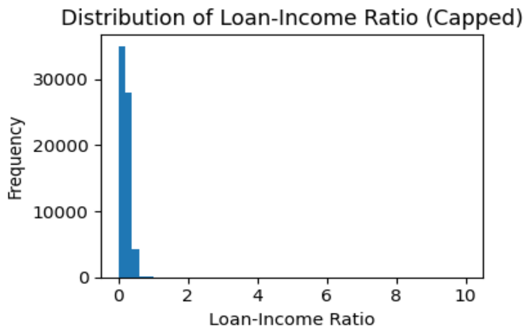

Figure 9: Distribution of the loan-to-income ratio after applying plausibility caps.

3.5 Correlation Analysis and Redundant Feature Removal

To further reduce redundancy among predictors, a correlation analysis was conducted on the numerical variables in the dataset. Only feature-feature correlations were examined, while the target variables (label and int_rate) were excluded to avoid target leakage. The Pearson correlation matrix was computed for 78 numerical features using a subset of **9,444** observations with complete values, since correlation estimation requires non-missing values across all variables. Feature pairs with very strong linear relationships were identified using a threshold of $|r| \geq 0.90$. **Figure 10** presents the correlation heatmap for the top **10** features, highlighting strongly related predictors. Several feature pairs exhibited extremely high correlations, indicating potential redundancy. Examples include **loan_amnt and funded_amnt (r = 1.00), open_acc and num_sats (r = 0.998)** etc. Since highly correlated variables provide similar information, retaining all of them may introduce multicollinearity and reduce model interpretability. Therefore, one variable from each highly correlated pair was removed, resulting in the removal of **10** redundant features. After this step, the dataset contained **80** columns, including both predictors and target variables.

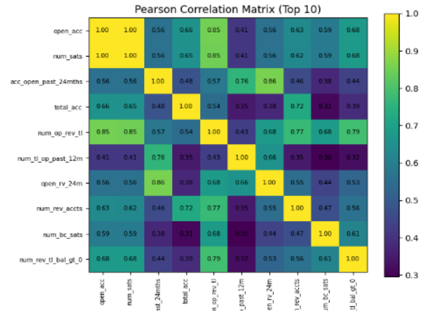

Figure 10: Top 10 features correlation heatmap.

3.6 Time-Aware Train-Test Split

To ensure a realistic evaluation of model performance, the dataset was divided into training and testing sets **using a chronological split based on the loan issue date.** In credit risk modeling, models are trained on historical data and applied to future loans; therefore, random splitting could introduce information from future observations into the training set and lead to overly optimistic results. Loans were ordered by the issue_date variable and the earliest **80%** of observations were assigned to the training set while the most recent **20%** formed the test set. This split produced **1,078,479** training observations and **269,620** testing observations, covering loans issued between **June 2007-October 2016** for training and **October 2016-December 2018** for testing. After splitting, the variables issue_date and id were removed because they were used only for ordering and identification and were not intended as predictive features. **Figure 11** shows the distribution of the binary default label in the training and testing sets, indicating that the class proportions remain similar across both datasets. **Figure 12** presents the distribution of interest rates, showing comparable patterns between the training and testing data.

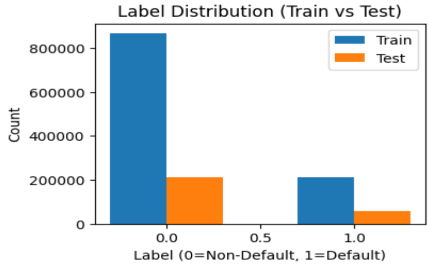

Figure 11: Label distribution.

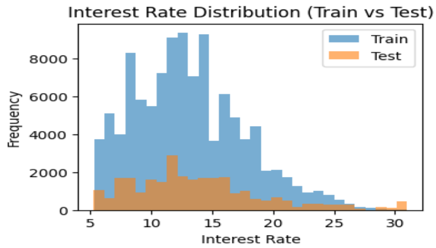

Figure 12: Interest rate distribution.

3.7 Missing Value Imputation

After performing the chronological train-test split, the remaining missing values were handled using a train-fitted imputation strategy, where imputation parameters are estimated using only the training dataset and then applied to both training and testing data. This approach prevents information from the test set influencing preprocessing decisions and avoids data leakage.

Inspection of the training data showed that **64** numerical feature columns contained missing values, while no categorical variables contained missing entries after earlier preprocessing steps. Therefore, the imputation procedure focused on numerical variables.

For most continuous features, missing values were replaced using median imputation computed from the training dataset. Median imputation was chosen because financial variables often exhibit skewed distributions, making the median more robust to extreme values than the mean.

A different strategy was applied to variables beginning with mths_since_, which represent the number of months since specific credit events occurred. In these variables, missing values typically indicate that the event has never occurred. To preserve this interpretation, missing entries were replaced with a sentinel value equal to the maximum observed value plus one. After computing imputation values on the training data, the same values were applied to both datasets. As a result, all missing values were removed, producing complete feature matrices suitable for machine learning models. After preprocessing, the dataset contains **1,348,099** observations and **78** feature columns (**70 numerical and 8 categorical variables),** along with the two target variables used for the learning tasks (label for classification and int_rate for regression). The complete dataset schema, including variable types, is provided in **Appendix A, Table A.1** for reproducibility. Further transformations such as categorical encoding, feature vector assembly, feature selection, and feature scaling will be performed within the modeling pipelines. These steps are task-specific and will be applied separately for loan default classification, interest rate regression, and borrower risk segmentation through clustering.

4 Classification Task

This part of the study addresses the research question: **Can we predict whether a loan will default?** The problem is formulated as a binary classification task where the model distinguishes between loans that were fully repaid and those that resulted in default. Predicting loan default is important for financial institutions because it helps assess borrower risk and improve credit decision-making. To evaluate predictive performance, two classification algorithms were applied: **Logistic Regression** and **Gradient-Boosted Trees (GBT)**. **Logistic Regression** was selected as a linear baseline model that provides interpretable relationships between predictors and the probability of default, while **Gradient-Boosted Trees** were used to capture more complex non-linear relationships and interactions between borrower characteristics, credit history variables, and loan attributes. Comparing these models allows us to assess whether loan default risk can be explained through a linear decision boundary or whether more flexible ensemble methods provide improved predictive performance.

4.1 Modeling Pipeline

A unified **PySpark** modeling pipeline was constructed to prepare the dataset for classification. After removing the target variable (label) and excluding **int_rate**, the dataset contained **68 numerical features and 8 categorical variables**. The interest rate variable was excluded because it may directly reflect lender risk assessment and could introduce information leakage into the default prediction task.

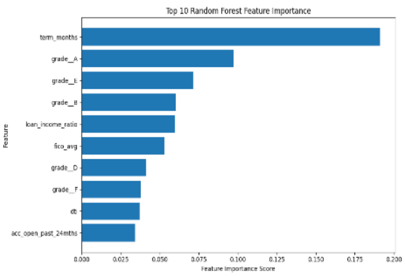

Figure 13: Top 10 Random Forest feature importance scores.

Categorical attributes were converted into numerical indices using **StringIndexer** fitted on the training dataset with **handleInvalid="keep"**, ensuring consistent category handling between training and test data. These indexed variables were then transformed using **OneHotEncoder** with **dropLast=True**, expanding categorical variables into binary indicators.

The encoded categorical features were combined with the numerical variables using **VectorAssembler**, resulting in a **163-dimensional feature vector** for both training and test datasets. Feature selection was performed using **Random Forest feature importance** computed on the training data. Features with an **importance score ≥ 0.002** were retained, reducing the feature space from **163 to 50 features**.

The selected features were extracted using **VectorSlicer** and used as the final input representation for model training. The feature importance distribution is shown in **Figure 13**, highlighting the **top 10 features** contributing to default prediction. Feature scaling was applied when training **Logistic Regression**, whereas **Gradient-Boosted Trees** were trained without scaling since tree-based algorithms are invariant to feature scale.

4.2 Logistic Regression

Logistic Regression was trained using the selected feature set and the predictors were standardized using **StandardScaler** prior to model fitting. Standardization ensures that all predictors contribute on a comparable scale and prevents variables with larger magnitudes from dominating the optimization process. Feature scaling was applied because Logistic Regression is a linear optimization-based model and differences in feature magnitude can affect coefficient estimation and convergence. In addition, scaling often improves numerical stability and helps the optimization algorithm converge more efficiently. Hyperparameter tuning was conducted using **3-fold cross-validation** on a stratified **20%** subset of the training data, with ROC-AUC used as the model selection metric. Stratified sampling was used to preserve the proportion of defaulted and non-defaulted loans during the tuning process, ensuring that the validation results remain representative of the original class distribution. The tuning process explored different levels of regularization strength, elastic-net mixing, iteration limits, and classification thresholds. Regularization parameters were included in the search to control model complexity and reduce the risk of overfitting, particularly given the relatively high dimensionality of the feature space. Two variants of the model were trained: an unweighted Logistic Regression model and a class-weighted version designed to compensate for the class imbalance between defaulted and non-defaulted loans. Class weighting increases the penalty associated with misclassifying the minority default class, which can improve the model's ability to detect risky borrowers.

The ROC curves for both models are shown in **Figure 14**, while the corresponding confusion matrices are presented in **Figure 15**. Although both variants produced very similar ROC performance, their classification

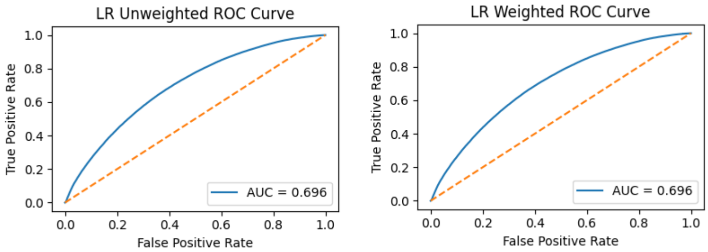

Figure 14: ROC curves for weighted and unweighted Logistic Regression.

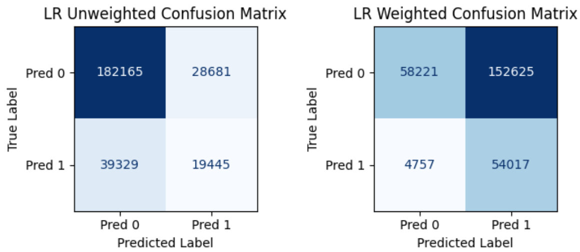

Figure 15: Confusion matrices for weighted and unweighted Logistic Regression.

behavior differs considerably. The unweighted produces a more balanced prediction outcome and substantially higher overall accuracy, whereas weighted model shifts predictions strongly toward the minority default class. This behavior is expected when class weights are applied, as the model prioritizes detecting defaults at the expense of increased false positive predictions. This increases recall for defaulted loans but results in many false positives and a large drop in overall accuracy. Such a trade-off is common in imbalanced classification problems, where improving detection of rare events often leads to reduced precision or overall accuracy. Considering the overall performance and the trade-off between precision and recall, the unweighted Logistic Regression model was selected as the preferred variant for subsequent comparison with Gradient Boosted Trees. This choice reflects a balance between predictive performance and practical usability, as extremely high false positive rates could lead to unnecessary rejection of low-risk borrowers. However, it is important to note that the weighted version significantly improves recall for defaults. So, if the objective were purely risk minimization, the weighted model could also be preferred. Detailed parameter settings and performance metrics for both models are summarized in **Table 3**.

Table 3: Logistic Regression hyperparameter tuning and performance

Model	Parameter grid explored	Best parameters	Accuracy	ROC-AUC	PR-AUC	Precision (Default) (Non-Default)	Recall (Default) (Non-Default)	F1-score (Default) (Non-Default)
LR Unweighted	regParam {0, 0.001, 0.01, 0.05, 0.1}; elasticNetParam {0, 0.25, 0.5, 0.75, 1}; maxIter {30, 50, 80, 120}; threshold {0.35, 0.40, 0.45, 0.50}	regParam=0.0, elasticNetParam=0.75, maxIter=50, threshold=0.35	0.7478	0.6960	0.3642	0.4040 0.8224	0.3308 0.8640	0.3638 0.8427
LR Weighted			0.4163	0.6955	0.3625	0.2614 0.9245	0.9191 0.2761	0.4070 0.4252

4.3 Gradient Boosted Trees

Gradient Boosted Trees (GBT) was trained on the same **50-feature** input used for Logistic Regression. Using the same feature set ensures that performance differences between the models reflect the modeling approach rather than differences in the input representation. Unlike Logistic Regression, feature scaling was not applied because tree-based models are invariant to monotonic changes in feature scale and do not depend on distance-based optimization. Tree-based models split observations based on feature thresholds rather than distances, which makes them robust to differences in feature magnitude. Hyperparameter tuning was conducted using **3-fold cross-validation** on a stratified **20%** subset of the training data, with ROC-AUC used as the model selection criterion. Stratified sampling ensures that both defaulted and non-defaulted loans are proportionally represented during the tuning process, preserving the original class distribution. The tuning procedure explored different combinations of tree depth, number of boosting iterations, learning rate, subsampling rate, and classification threshold in order to identify the configuration that provided the best predictive performance. These parameters control the complexity and learning behavior of the ensemble, where deeper trees increase model flexibility, learning rate determines the contribution of

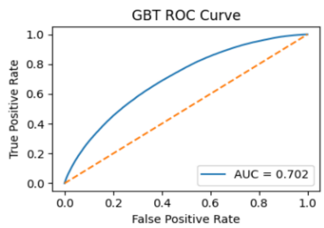

Figure 16: ROC curve of Gradient Boosted Trees.

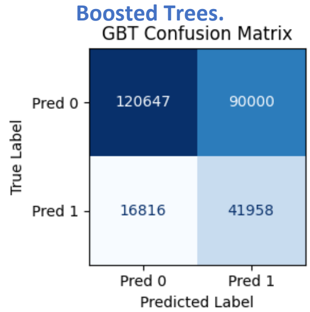

Figure 17: Confusion matrix of Gradient Boosted Trees.

each tree, and subsampling can help reduce overfitting. The ROC curve and confusion matrix are shown in **Figure 16** and **Figure 17**, respectively, while the complete tuning range and final test performance are summarized in **Table 4**. Feature importance scores extracted from the model further highlight the predictors that contribute most strongly to the classification task, illustrated in **Figure 18**. Compared with Logistic Regression, GBT achieved a higher ROC-AUC and PR-AUC, indicating better ranking ability and stronger discrimination between defaulted and no n-defaulted loans. It also produced substantially higher recall for the default class than the unweighted Logistic Regression model, although this came at the cost of lower overall accuracy. This suggests that GBT is more effective at identifying risky borrowers, while Logistic Regression remains more conservative and accurate overall. If the primary objective is to detect as many defaulted loans as possible, GBT provides a stronger classification strategy. Therefore, the choice between the two models ultimately depends on the practical objective of the lending institution, whether prioritizing accurate overall classification or maximizing detection of potential defaults.

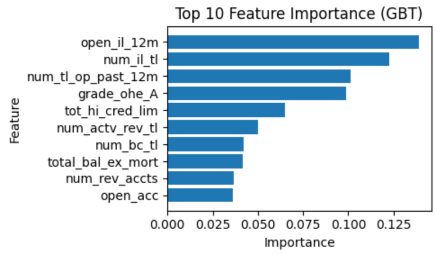

Figure 18: Top 10 Feature Importance (GBT)

Table 4: GBT hyperparameter tuning and performance

Model	Parameter grid explored	Best parameters	Accuracy	ROC-AUC	PR-AUC	Precision (Default) (Non-Default)	Recall (Default) (Non-Default)	F1-score (Default) (Non-Default)
GBT	maxDepth {3, 5, 7, 9}; maxIter {20, 40, 60, 80}; stepSize {0.05, 0.1, 0.2}; subsamplingRate {0.6, 0.8, 1.0}; threshold {0.20, 0.25, 0.30, 0.35, 0.40, 0.50}	maxDepth=5, maxIter=40, stepSize=0.1, subsamplingRate=1.0, threshold=0.20	0.6031	0.7025	0.3837	0.3175 0.8777	0.7139 0.5722	0.4395 0.6928

4.4 Model Comparison and Conclusion

Model performance was primarily evaluated using recall and F1-score for the default class, along with PR-AUC and ROC-AUC, as the dataset is imbalanced and correctly identifying defaulted loans is more important than maximizing overall accuracy. These metrics provide a more informative assessment of classifier performance in imbalanced settings, where traditional accuracy can be misleading due to the dominance of the majority class. The experimental results show that both models can predict loan default using borrower financial and credit characteristics, but their performance differs depending on the evaluation objective. Logistic Regression achieved higher overall **accuracy (0.7478)** and produced more balanced predictions between defaulted and non-defaulted loans. This behavior reflects the more conservative nature of linear models, which tend to avoid excessive positive predictions for the minority class. However, Gradient Boosted Trees achieved stronger ranking performance, with higher **ROC-AUC (0.7025 vs. 0.6960)** and **PR-AUC (0.3837 vs. 0.3642).** Higher PR-AUC is particularly important in this context because it reflects the model's ability to correctly identify defaulted loans among all predicted high-risk cases. More importantly, GBT significantly improved the detection of risky borrowers, achieving a **default recall of 0.7139** compared to **0.3308 for Logistic Regression**. This indicates that the tree-based model is better at identifying loans that are likely to default, which is a critical objective in credit risk modeling. Capturing a larger proportion of potential defaults can help financial institutions take preventive actions such as adjusting credit limits, increasing monitoring, or modifying lending decisions. Although GBT produces lower overall accuracy due to an increased number of false positives, the improved ability to detect high-risk borrowers makes it more suitable for financial risk assessment. In practical lending scenarios, accepting a higher number of false alarms may be preferable to failing to detect borrowers who are likely to default. Therefore, **Gradient Boosted Trees (GBT) is the best-performing model for the loan default prediction task in this study**, as it provides stronger discrimination and substantially better identification of defaulted loans. These findings suggest that nonlinear relationships and complex interactions between borrower characteristics play an important role in explaining default risk, which are better captured by ensemble tree-based methods than by linear models.

5 Regression Task

The prediction task is formulated as a supervised regression problem where the goal is to estimate the continuous target variable, the loan interest rate, using borrower financial characteristics and loan attributes. This study addresses the research question: **Can borrower financial characteristics and loan attributes be used to predict loan interest rates?**

The regression task aims to estimate the interest rate assigned to each loan. Interest rates are influenced by multiple borrower-related factors and loan characteristics. To investigate this problem, regression models were evaluated using the Lending Club dataset. Three regression models were used to predict loan interest rates from borrower characteristics.

First, a standard **Linear Regression model** was applied to capture the linear relationship between borrower financial indicators and interest rates. To investigate whether emphasizing higher-interest loans could improve prediction performance, a **weighted Linear Regression model** was also trained. Finally, a **Gradient Boosted Trees (GBT)** model was implemented as a nonlinear ensemble method capable of capturing more complex interactions and nonlinear patterns between borrower attributes and loan pricing.

5.1 Modeling Pipeline

A PySpark modeling pipeline was constructed to prepare the dataset for regression analysis. **Categorical variables** were removed from the dataset so that the regression models could be trained only on numerical predictors. In addition, the variables **grade** and **label** were excluded to prevent potential information leakage. After removing categorical attributes, the dataset contained only numerical variables, with **interest rate (int rate)** as the target. Feature selection was then performed using **Random Forest** feature importance to reduce dimensionality, as illustrated in **Figure 19**. Random Forest feature importance was used for feature selection because tree-based models can capture nonlinear relationships and interactions between variables. A Random Forest regressor was trained on a **20%** stratified sample of the training data to reduce computational cost while still preserving the overall data distribution. The importance of each predictor was extracted and features were ranked according to their contribution to prediction accuracy. The **top 20 numerical variables** were then selected as the final feature set for model training, reducing noise while retaining the most informative predictors.

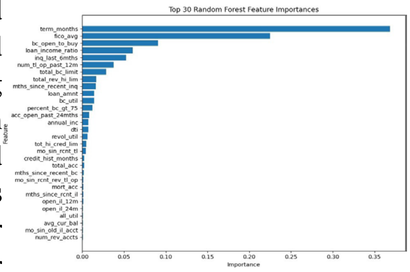

Figure 19: Random Forest Feature Importance used for selecting the top predictive variables.

These selected predictors represent borrower credit characteristics and financial capacity indicators and were combined into a single feature vector using **Vector Assembler** to create a structured input suitable for spark machine learning models. Feature scaling using **Standard Scaling** was applied for **linear regression** to normalize feature magnitude and stabilize coefficients estimation, while **Gradient Boosted Trees** were trained **without scaling** since tree-based models are invariant to feature scaling.

5.2 Linear Regression

Linear Regression was applied to model the relationship between the selected borrower financial indicators and the loan interest rate. **Hyperparameter tuning** was performed using **3-fold cross-validation** on the training data. The parameter grid included multiple values of the regularization parameter (regParam), elastic net mixing parameter (elasticNet Param), and maximum number of optimization iterations (maxIter). Root Mean Squared Error (RMSE) was used as the model selection metric during cross-validation.

Two variants of Linear Regression were trained. The first model represents the standard **unweighted Linear Regression**, where all observations contribute equally to the training process. The second model is a **weighted Linear Regression,** where a weight column proportional to the observed interest rate was introduced. Higher interest rates typically correspond to riskier loans, which often contain stronger signals about borrower credit risk and loan pricing. Assigning larger weights to observations with higher interest rates encourages the model to place greater emphasis on accurately predicting these high-risk loans during training.

The hyperparameter tuning results for both models are summarized in **Table 5**. For both the unweighted and weighted variants, the best configuration corresponded to (**regParam = 0.0, elasticNetParam = 0.0)** with a maximum of **30 iterations**.

Table 5: Linear Regression hyperparameter tuning and performance

Model	Parameter grid explored	Best parameters	RMSE	MAE	R^2
LR Unweighted	regParam {0, 0.001, 0.01,0.05,0.1} elasticNetParam {0, 0.25, 0.5,0.75, 1} maxIter {30, 50, 80}	regParam = 0.0 elasticNetParam = 0.0 maxIter = 30	4.5778	3.2229	0.314
LR Weighted	regParam {0, 0.001, 0.01, 0.05,0.1} elasticNetParam {0, 0.25, 0.5, 0.75, 1} maxIter {30, 50, 80}	regParam = 0.0 elasticNetParam = 0.0 maxIter = 30	4.4501	3.2508	0.352

The predictive performance of the models was evaluated on the test dataset using **RMSE, MAE, and R2**, with the detailed results reported in **Table 5**. **The weighted Linear Regression** shows a slight improvement over the standard Linear Regression, indicating that emphasizing higher- interest loans during training allows the model to capture additional variation in interest rate prediction. The relationship between predicted and actual interest rates is shown in **Figure 20**. Predictions generally follow the diagonal line, indicating that the models capture the overall trend, although linear models tend to underestimate very high interest rates. **The Residual vs Predicted plots** show residuals centered around zero, indicating no strong systematic bias. However, the spread of residuals increases for larger predicted values, indicating heteroscedasticity, where prediction errors tend to grow for loans with higher interest rates. This pattern suggests that prediction errors increase for loans with higher interest rates, indicating that the relationship between borrower characteristics and interest rates may become more complex for higher-risk loans. Because the predictors were standardized before training, the regression coefficients represent the effect of a one standard deviation increase in each variable rather than a one-unit change. Regression coefficients were interpreted based on their absolute magnitude. The most influential predictors include loan term **(term months, 1.74)**, average credit score **(fico avg, -1.27)**, total credit limit **(total bc limit, -0.73)**, recently opened credit accounts **(num tl op past 12m, 0.73)**, and recent credit inquiries **(inq last 6mths, 0.67).**
Additional factors such as the loan-to-income ratio (loan income ratio, 0.65) and high credit card utilization (percent bc gt 75, 0.60) also contribute to interest rate variation. Overall, the Linear Regression models explain part of the interest rate variation, but the results suggest that purely linear relationships cannot fully capture the complexity of loan pricing.

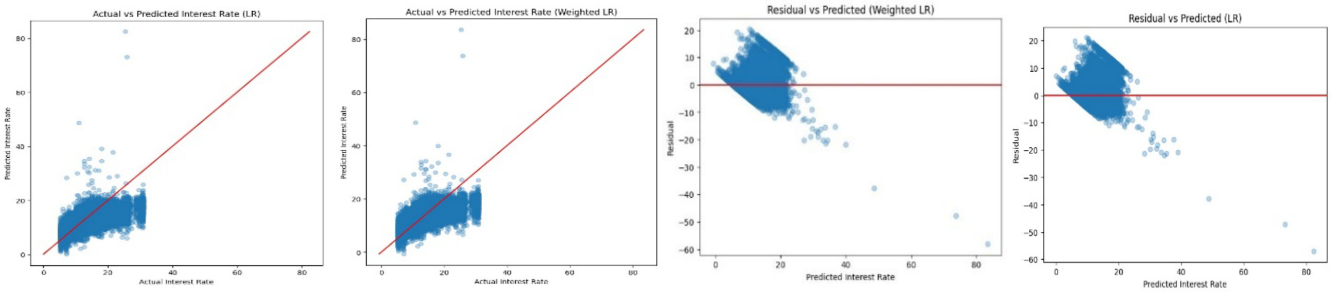

Figure 20: Actual vs Predicted plots and Residual vs Predicted plots for unweighted & weighted Linear Regression models.

5.3 Gradient Boosted Trees

Gradient Boosted Trees (GBT) were used as a **nonlinear regression model** to capture more complex relationships between borrower characteristics and loan interest rates. Gradient Boosted Trees build sequential decision trees to capture nonlinear relationships, and no feature scaling is required for tree- based models. Hyperparameter tuning was performed using 3-fold cross- validation on a **20%** subset of the training data to reduce computational cost. The parameter grid tested different combinations of parameters with results reported in **Table 6.**

Table 6: Gradient Boosted Trees hyperparameter tuning and performance

Model	Parameter grid explored	Best parameters	RMSE	MAE	R2
GBT	maxDepth {3, 5, 7} maxIter {40, 80} stepSize {0.05, 0.1} subsamplingRate {0.8, 1.0}	maxDepth = 5 maxIter = 80 stepSize = 0.1 subsamplingRate = 1.0	4.3304	3.1230	0.3865

The predictive performance of the model is illustrated in **Figure 21**, which shows the **Actual vs Predicted interest rate plot**. Most predictions follow the diagonal line representing perfect prediction, indicating that the model captures the overall pattern of interest rate variation more effectively than the linear models. The Residual vs Predicted plot **Figure 21** shows residuals centered around zero with a relatively stable spread, suggesting that the model reduces systematic prediction errors compared to the linear models. Feature importance scores extracted from the model further highlight the predictors that contribute most strongly to interest rate estimation.

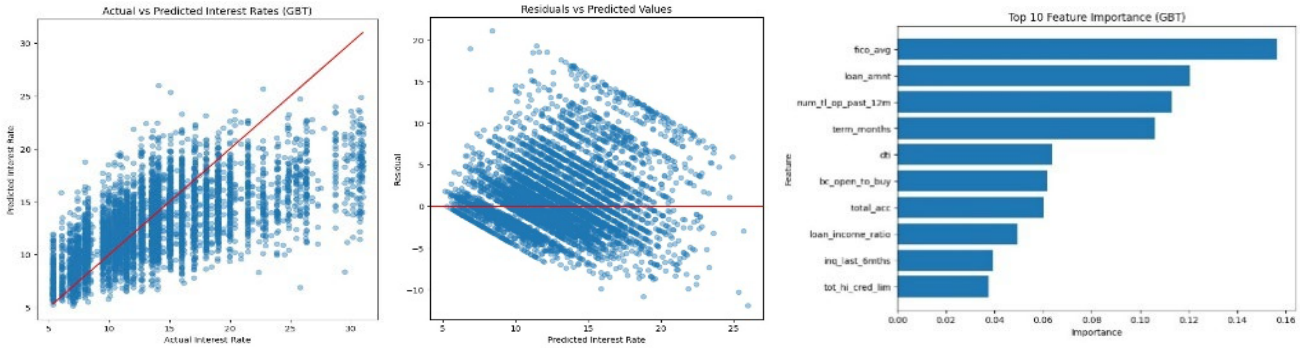

Figure 21: Diagnostic plots and feature importance for the Gradient Boosted Trees model.

5.4 Model Comparison and Conclusion

This study investigated whether borrower financial characteristics and loan attributes can be used to predict loan interest rates using regression models. The results show that these variables contain meaningful predictive information about loan pricing.

The experimental results show that both **Linear Regression** and **Gradient Boosted Trees** are able to capture meaningful relationships between borrower financial characteristics and loan interest rates. The weighted Linear Regression slightly improves performance over the standard Linear Regression, reducing **RMSE** from **4.5778 to 4.4501** and increasing R^2 from **0.314 to 0.352**. This improvement suggests that assigning higher importance to loans with larger interest rates helps the model better represent patterns associated with riskier borrowers.

However, the **Gradient Boosted Trees (GBT) model** achieved the **best overall** predictive performance. Compared with the best Linear Regression model, GBT further reduced the RMSE to **4.3304** and increased the R^2 to **0.3865**, indicating that the nonlinear model captures additional variation in loan interest rates that cannot be explained by linear relationships alone.

Overall, the findings suggest that while borrower financial indicators explain a substantial portion of interest rate variation, more flexible nonlinear models such as Gradient Boosted Trees provide more accurate predictions and are better suited for modeling the complex structure of loan pricing.

6 Clustering Task

This section investigates whether borrowers can be grouped into **natural risk segments** using unsupervised learning. Unlike supervised models that rely on predefined outcome labels, clustering can reveal hidden structures in borrower behavior by grouping observations with similar financial characteristics. Clustering techniques, including **K-Means and Bisecting K-Means**, were therefore applied to identify groups of borrowers with similar credit behavior and financial profiles within the LendingClub dataset.

6.1 Selection and Preparation

To perform borrower segmentation, clustering was applied to a subset of behavioral financial variables selected from the preprocessed dataset. The features used for clustering were **revol_util, percent_bc_gt_75, dti, inq_last_6mths, acc_open_past_24mths, pct_tl_nvr_dlq, num_actv_rev_tl, and credit_hist_months**. These variables were selected because they capture complementary aspects of borrower credit behavior, including **credit utilization, borrowing activity, repayment reliability, and the length of credit history**, which are key indicators of financial risk.

Because clustering algorithms such as K-Means rely on **distance calculations**, the feature set was limited to variables that provide distinct behavioral information while avoiding redundant or highly correlated predictors. The credit history length variable was **log-transformed** to reduce skewness, resulting in the feature *log_credit_hist_months* used in the clustering analysis. In addition, numerical features were **winsorized using the 1st and 99th percentile thresholds** to reduce the influence of extreme values. Finally, all variables were **standardized** before clustering. This step is necessary because distance-based algorithms are sensitive to feature scale, and variables with larger numeric ranges could otherwise dominate the clustering process.

6.2 KMeans Clustering

To determine an appropriate number of clusters, K-Means models were evaluated for values of **K ranging from 2 to 11**. K-Means partitions observations into **K clusters** by iteratively assigning each observation to the nearest centroid and updating cluster centers in order to minimize the **within-cluster sum of squared distances** between observations and their assigned centroids. Two evaluation metrics were used to select the appropriate number of clusters: the **within-cluster sum of squared errors (SSE)** and the **silhouette score.**

Figure 22 shows the SSE values for different numbers of clusters. The within-cluster sum of squared errors (SSE) measures the total squared distance between observations and their assigned cluster centroids, with lower values indicating more compact clusters. As expected, SSE decreases as the number of clusters increases, since adding clusters reduces the average distance between observations and their assigned centroids. However, the rate of improvement begins to diminish after **k = 3**, indicating an elbow-like pattern where additional clusters provide only marginal reduction in SSE. For example, SSE decreases substantially from $\mathbf{8.98 \times 10^6}$ at **k = 2** to $\mathbf{7.95 \times 10^6}$ at **k = 3**, while subsequent reductions become progressively smaller.

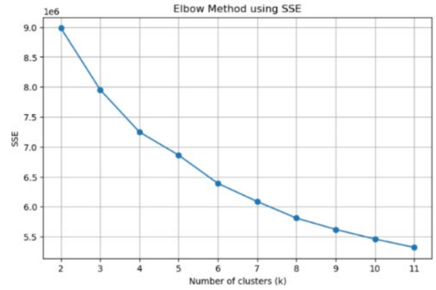

Figure 22: Elbow Curve for K-Means Clustering (SSE vs Number of Clusters)

Figure 23 presents the silhouette scores obtained for the same models. The silhouette score ranges from **−1 to 1**, where higher values indicate better separation between clusters and stronger internal cohesion. The highest silhouette score occurs at **k = 4 (0.278)**; however, two clusters would merge distinct borrower behaviors into a single group, while the score for **k = 3 (0.271)** remains very close. Since silhouette values measure the separation between clusters relative to their internal cohesion, this result suggests that both **k = 3** and **k = 4** provide reasonable clustering structures.

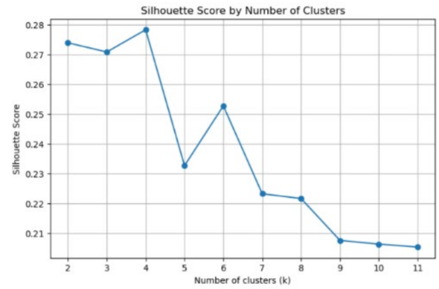

Figure 23: Silhouette Score for K-Means Across Different k Values

Considering the elbow pattern in SSE, the comparable silhouette performance between **k = 3** and **k = 4**, and the clearer separation observed in the PCA visualization, the final K-Means model was trained using k = 3 clusters.

The clustering structure obtained using the K-Means model is visualized in the PCA projection shown in **Figure 24**. The projection displays three distinct clusters in the two-dimensional space defined by the first two principal components. These two components explain approximately **43.8%** of the total variance in the dataset, allowing the high-dimensional borrower features to be represented in a reduced two-dimensional space for visualization purposes only, while clustering itself was performed on the original standardized features. The visualization indicates that the three-cluster solution produces clearly distinguishable regions with relatively limited overlap between clusters, whereas increasing the number of clusters leads to more fragmented groups and greater overlap in the projected space.

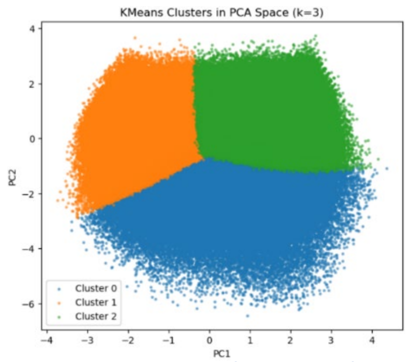

Figure 24: K-Means Cluster Visualization in PCA Projection (k = 3)

The characteristics of the resulting clusters are summarized in **Table 7**, revealing three distinct borrower behavior profiles.

Table 7: K-Means Cluster Characteristics (k = 3)

Cluster	Size	Avg FICO	Avg Interest (%)	Utilization (%)	% BC >75%	DTI	Accounts Open (24m)	Inquiries (6m)
0	304,547	692.1	**14.54**	46.6	37.7	**22.37**	**7.93**	**1.33**
1	542,537	687.5	14.03	**72.47**	**75.93**	19.44	3.35	0.42
2	501,015	**713.46**	**11.60**	**32.59**	**17.00**	**14.12**	4.04	0.50

Cluster 2 (Financially stable borrowers) represents the most financially stable group, with the highest average FICO score **(713.46),** the lowest interest rate **(11.60%),** and relatively low credit utilization and debt-to-income ratio. In contrast, **Cluster 1 (High credit-utilization borrowers)** is characterized by very high credit utilization **(72.47%)** and a large proportion of bankcards used above **75%** of their limit, indicating significant credit exposure despite a slightly lower average FICO score. **Cluster 0 (Credit-seeking borrowers)** exhibits a different risk pattern: although its utilization is moderate, borrowers in this cluster open substantially more credit accounts within the previous **24 months (7.93 on average)** and show the highest inquiry activity, suggesting aggressive credit-seeking behavior. This indicates that the clusters do not simply represent increasing levels of risk, but rather capture different behavioral pathways through which borrower risk can emerge.

6.3 Bisecting K-Means

To further examine the behavioral structure of borrowers, the Bisecting K-Means (BKM) algorithm was applied using the same eight behavioral features used in the standard K-Means model. Bisecting K-Means is a hierarchical variant of K-Means that recursively splits clusters into two subclusters, allowing the algorithm to progressively divide the data into more homogeneous groups.

The choice of **k = 3** is supported by the evaluation metrics. As illustrated in Figure 25, the Silhouette curve reaches its maximum at k = 2 and gradually decreases afterward, whereas the SSE curve shown in Figure 26 exhibits a clear elbow **around k = 3-4**, where the marginal reduction in SSE begins to diminish. Although two clusters would provide the highest compactness according to the Silhouette score, such a configuration would oversimplify borrower behavior. Therefore, **k = 3 was selected** as a balanced solution, allowing the model to capture distinct behavioral patterns while maintaining reasonable cluster separation.

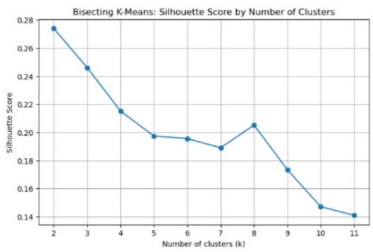

Figure 25: Silhouette Score for BK-Means Across Different k Values

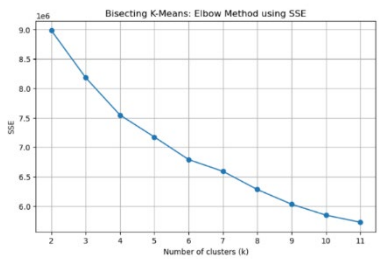

Figure 26: Elbow Curve for BK-Means Clustering (SSE vs Number of Clusters)

For **k = 3**, the Bisecting K-Means model achieved an SSE of **8.18 × 10^6** and a Silhouette score of **0.246**. While this silhouette value is slightly lower than that obtained with standard K-Means, the hierarchical splitting strategy enables the identification of clusters that reflect meaningful differences in borrower financial behavior.

The clustering structure obtained using the Bisecting K-Means model is visualized in the PCA projection shown in **Figure 27**. The projection displays three distinct clusters in the two-dimensional space defined by the first two principal components. Despite this dimensionality reduction, the projection reveals clearly distinguishable regions corresponding to the three clusters, indicating that the Bisecting K-Means algorithm captures meaningful patterns in borrower financial behavior.

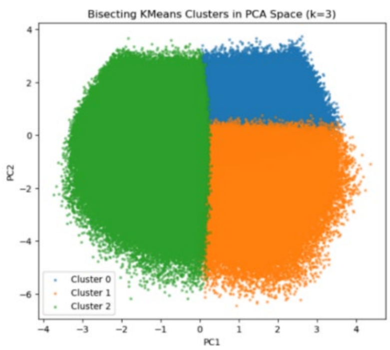

Figure 27: BK-Means Cluster Visualization in PCA Projection (k = 3)

The characteristics of the resulting Bisecting K-Means clusters are summarized in **Table 8**. The segmentation again reveals three distinct borrower behavior profiles. **Cluster 0 (Financially stable borrowers)** represents the most financially stable group, with the highest average FICO score (714.5), the lowest average interest rate (11.42%), and relatively low credit utilization and debt-to-income ratio.

Cluster 2 (High credit-utilization borrowers) is characterized by very high credit utilization **(70.41%)** and a high proportion of bankcards used above 75% of their limit, indicating substantial credit exposure; this group also has the lowest average FICO score (686.85) and the highest average interest rate (14.21%). **Cluster 1 (Credit-seeking borrowers)** reflects a different behavioral pattern: although its utilization is much lower than that of Cluster 2, borrowers in this group open substantially more accounts within the previous **24 months (8.18 on average)** and show the highest inquiry activity, suggesting intensive credit-seeking behavior. These results indicate that the Bisecting K-Means model separates borrowers not only by overall risk level, but also by distinct behavioral patterns through which financial risk may emerge.

Table 8: BK-Means Cluster Characteristics (k = 3)

Cluster	Size	Avg FICO	Avg Interest (%)	Utilization (%)	% BC >75%	DTI	Accounts Open (24m)	Inquiries (6m)
0	439,240	**714.5**	**11.42**	32.41	16.47	14.04	3.79	0.41
1	240,133	699.77	13.87	35.42	20.74	19.99	**8.18**	**1.46**
2	668,726	686.85	**14.21**	**70.41**	**73.23**	20.14	3.92	0.53

6.4 Model Comparison and Conclusion

The clustering analysis shows that borrower behavior in the dataset is not defined by a single linear risk scale, but by distinct behavioral patterns. Across both clustering approaches, three consistent borrower profiles emerge: financially stable borrowers with strong credit scores and low utilization, borrowers with very high credit utilization and substantial credit exposure, and borrowers with intensive credit-seeking behavior characterized by frequent account openings and higher inquiry activity.
These findings suggest that borrower risk is influenced not only by credit scores but also by behavioral mechanisms related to credit usage and borrowing activity.

Both K-Means and Bisecting K-Means identify broadly similar behavioral profiles within the borrower population. However, the cluster structures differ: K-Means produces slightly more compact clusters, while Bisecting K-Means, through its hierarchical splitting strategy, generates clusters with different sizes and boundaries. Overall, the results demonstrate that unsupervised clustering can reveal meaningful borrower segments and provide additional insights into credit usage patterns beyond traditional risk indicators.

7 Conclusion with Limitations and Future Work

This project explored borrower characteristics and credit risk patterns using the **LendingClub dataset** through data mining techniques such as classification, regression, and clustering.

The analysis began with **exploratory data analysis and preprocessing steps** to prepare the dataset for machine learning tasks. Missing values, redundant features, and potential data leakage variables were handled, and new features were engineered to better represent borrower financial behavior.

For the **classification task**, the objective was to predict whether a loan would default. Logistic Regression and Gradient Boosted Trees were evaluated for this purpose. Logistic Regression achieved higher overall accuracy, while Gradient Boosted Trees showed a stronger ability to detect defaulted loans due to its capability to model nonlinear relationships. For **regression tasks**, Gradient Boosted Trees also outperformed Linear Regression when predicting loan interest rates. **Clustering techniques** revealed three distinct borrower behavior groups, showing that borrower risk can arise from different behavioral patterns such as high credit utilization or intensive credit-seeking activity.

However, this study has some limitations. The models rely on historical LendingClub data, which may not fully generalize to other lending platforms or economic conditions, and the dataset contains class imbalance that makes default prediction more challenging.

Future work could explore more advanced models such as XGBoost or LightGBM, apply improved imbalance-handling techniques and integrate Explainable AI (XAI) methods to better understand model decisions while further enhancing predictive performance and credit risk analysis.

Appendix A: Final Dataset Schema

Table A.1: Schema of the loan dataset including column names, data types, and categories.

Column Name	Category	Data Type
loan_amnt	Numerical	Double
int_rate	Numerical	Double
grade	Categorical	String
home_ownership	Categorical	String
annual_inc	Numerical	Double
verification_status	Categorical	String
purpose	Categorical	String
addr_state	Categorical	String
dti	Numerical	Double
delinq_2yrs	Numerical	Double
inq_last_6mths	Numerical	Double
mths_since_last_delinq	Numerical	Double
mths_since_last_record	Numerical	Double
open_acc	Numerical	Double
pub_rec	Numerical	Double
revol_bal	Numerical	Double
revol_util	Numerical	Double
total_acc	Numerical	Double
initial_list_status	Categorical	String
collections_12_mths_ex_med	Numerical	Double
mths_since_last_major_derog	Numerical	Double
application_type	Categorical	String
acc_now_delinq	Numerical	Double
tot_coll_amt	Numerical	Double
open_acc_6m	Numerical	Double
open_act_il	Numerical	Double
open_il_12m	Numerical	Double
open_il_24m	Numerical	Double
mths_since_rcnt_il	Numerical	Double
il_util	Numerical	Double
open_rv_12m	Numerical	Double
open_rv_24m	Numerical	Double
max_bal_bc	Numerical	Double
all_util	Numerical	Double
total_rev_hi_lim	Numerical	Double
inq_fi	Numerical	Double
total_cu_tl	Numerical	Double
inq_last_12m	Numerical	Double
acc_open_past_24mths	Numerical	Double
avg_cur_bal	Numerical	Double
bc_open_to_buy	Numerical	Double
bc_util	Numerical	Double
chargeoff_within_12_mths	Numerical	Double
delinq_amnt	Numerical	Double
mo_sin_old_il_acct	Numerical	Double
mo_sin_rcnt_rev_tl_op	Numerical	Double
mo_sin_rcnt_tl	Numerical	Double
mort_acc	Numerical	Double
mths_since_recent_bc	Numerical	Double
mths_since_recent_bc_dlq	Numerical	Double
mths_since_recent_inq	Numerical	Double
mths_since_recent_revol_delinq	Numerical	Double
num_accts_ever_120_pd	Numerical	Double
num_actv_bc_tl	Numerical	Double
num_actv_rev_tl	Numerical	Double
num_bc_sats	Numerical	Double
num_bc_tl	Numerical	Double
num_il_tl	Numerical	Double
num_op_rev_tl	Numerical	Double

num_rev_accts	Numerical	Double
num_tl_120dpd_2m	Numerical	Double
num_tl_30dpd	Numerical	Double
num_tl_90g_dpd_24m	Numerical	Double
num_tl_op_past_12m	Numerical	Double
pct_tl_nvr_dlq	Numerical	Double
percent_bc_gt_75	Numerical	Double
pub_rec_bankruptcies	Numerical	Double
tax_liens	Numerical	Double
tot_hi_cred_lim	Numerical	Double
total_bal_ex_mort	Numerical	Double
total_bc_limit	Numerical	Double
disbursement_method	Categorical	String
label	Numerical	Integer
term_months	Numerical	Integer
emp_length_years	Numerical	Integer
credit_hist_months	Numerical	Double
fico_avg	Numerical	Double
loan_income_ratio	Numerical	Double